Amit Ghimire

SUMMARY

Combining 5+ years of Games/AR/VR design and development experience with graduate education in Human-Computer Interaction (HCI), I bring a strong foundation in Software Engineering and Human-Centered Design methods. I excel at creating high-performance games and VR/AR applications catering to the needs and comfort of the user. I am an avid learner aspiring to contribute to improving human-machine interfacing technologies, especially for interactions for Games, Head Mounted Displays, and 3D Interfaces.

CAREER INTERESTS

Virtual/Augmented/Mixed Reality, Game Design, Game Development, Human-Computer Interaction, Human-AI Interaction, 3D User Interfaces

SKILLS

- Software Development (.NET MAUI, C#, Android Development with Java)
- Game, AR/VR/MR and Real-time 2D/3D application Development (Unity, XR Interaction, Meta XR SDK, C#, C++, OpenGL)
- Unity Networking (Photon Unity Networking, Unity Netcode, Unity Gaming Services)
- Basic Web Development (HTML, CSS, Javascript)
- Game Design, User Interaction Design, User Experience Research, Qualitative and Quantitative Research Methods
- Project Management (Agile, SCRUM, Jira, Trello)
- Version Control (Git, Unity Collaborate)
- Data Analysis (Python, R)

WORK EXPERIENCE

September 2024 - Current

Blackbox Technologies -Senior Unity Developer

- Developing and optimizing 3D eCommerce experiences using Unity (WebGL), focusing on immersive product browsing, customizable product displays, and interactive user interfaces.
- Integrating Firestore as the backend for real-time product data, ensuring smooth synchronization between product databases and 3D storefronts.
- Building and maintaining Python automation scripts to process and transform product data from multiple ecommerce api into a standardized format for use in the 3D store.
- Troubleshooting performance issues related to WebGL rendering, asset loading, and memory usage, particularly for mobile browsers.

May 2022 - December 2023
University of British Columbia -Graduate Research Assistant

- Collaborated with Korea Institute of Science and Technology in Meta-Humanoid Project to combine humanoid robot and AR avatar to enhance Robot-Mediated Telepresence Experience
- Developed a Mixed Reality Simulation to conceptually prototype “Meta-Humanoid” and ran a mixed-method user study using this simulation for experimental evaluation of the concept
- Deliverables : Humanoid Robot Simulation framework, Meta-Humanoid System Design, User Study Design, User Study Data Analysis, User Study Report

September 2021 - December 2023
University of British Columbia -Graduate Teaching Assistant

- Introduction to Computation in Engineering Design (APSC 160), Internet Computing (CPSC 317)

January 2020 - August 2021
Danson Solutions - Game Developer and Designer

- Worked on an e-learning platform called Chimpvine to design and develop 2D, 3D, and XR K-12 educational games collaborating with teachers to translate course syllabi to games and gamification of learning contents[Chimpvine]
- Collaborate with data analysts to track student performance and behaviors by recording their decisions and interactions in virtual environments to make data-driven improvements in learning content, and educational game designs

January 2017 - December 2019
Paracosma, Kathmandu - Software and Usability Engineer

- Successfully designed, developed, and conducted usability tests for single/multi-user XR applications/training simulations complying with specific client requirements and requests
- Helped to increase development productivity by building a suite of automation tools and pipeline extending the unity editor’s functionality and reducing the application development timeline
- Analyze application resource utilization, identify performance bottlenecks, test and recommend alternatives to improve XR software performance

EDUCATION

September 2021 - May 2024
University of British Columbia, Vancouver - Masters, Computer Science (Human Computer Interaction)
Recipient of “Designing For People (DFP) - Collaborative Research and Training Experience Program (CREATE)” Grant

November 2012 - December 2016
Institute of Engineering, Pulchowk Campus - Bachelors, Electronics and Communication Engineering

PUBLICATIONS

Amit Ghimire, Anova Hou, Ig-Jae Kim, Dongwook Yoon (2025)

AvatARoid: A Motion-Mapped AR Overlay to Bridge the Embodiment Gap Between Robots and Teleoperators in Robot-Mediated Telepresence.

In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25), Article 906, pp. 1–26.*
https://doi.org/10.1145/3706598.3713812

PROJECTS

- ### Simulation for Human-Robot Telepresence Interactions in AR
  (***Unity3D, Robotics Simulation, Mixed Reality, Azure Kinect, Experimental Study***)
  - Simulate visually and behaviorally realistic humanoid robots in AR to explore robotic telepresence systems
  - Conducted user research using the simulation to compare interaction in a robotic telepresence system when the teleoperator is shown with a)just the robot, b) the robot and the teleoperator's video, and c) The robot superimposed with an AR avatar
  - User Research on the viability of such simulations for exploration studies
  - *Manuscript and demo vids under review for publication in an international journal*

- ### Surgical Simulation training solutions in VR
  (***Unity3D, Soft Body Simulation, Fluid Simulation, Virtual Reality***)
  - Collaborated with a team of 3D artists and designers to successfully build a VR training simulation platform for laparoscopic cholecystectomy surgery simulation providing physics-based simulation and interaction with organs and surgical tools
  - Analyze the performance of different CPU/GPU-based physics frameworks in VR for physics-based deformation of 3D models and soft body/fluid simulation
  - Research on the usability of such simulation for training applications
  - https://www.youtube.com/playlist?list=PLQ7N7ffrusU4HQLTsePnQmLfLQLsvgTUZ

- ### Collaboration Platform in VR
  (***Unity3D, Unity Networking, Inverse Kinematics, Virtual Reality***)
  - Developed a networked VR application enabled by voice communication and interaction with each other and networked objects(3D objects, documents, whiteboards, audio/video playback, etc) in shared space in VR
  - Evaluate and implement different networking solutions in unity3D VR for synchronization of user avatars and interactions
  - Study design requirements of Networked VR applications such as virtual space awareness, proxemic behaviors, and multimodal interactions in social VR
  - Git Repo, Demo Video

- Motion Capture System for VR
  (*Unity3D, Unity Animation, Inverse Kinematics, Kalman Filter, Virtual Reality, Augmented Reality*)
  - Assisted in character animation process by implementing low fidelity motion capture system using HTC VIVE HMD, controllers, and trackers to drive 3D models of humanoid characters from user motion data to quickly iterate and test character animations to be tested in VR, AR, or 3D applications
  - Git Repo

- Virtual Tourism
  (*Unity3D, Photogrammetry, Virtual Reality, Augmented Reality*)
  - Developed a VR tourism application that enables users to virtually travel to different religious pilgrimages in Nepal with 360 videos of the site landmarks and 3D digital twin of pilgrimages created by photogrammetry artists and videographers
  - Created a virtual 3D agent as a travel guide to provide cultural, historical, and other information about the landmarks
  - https://www.youtube.com/watch?v=eg8MNS-czJ0

- Poly AR
  (*Unity3D, Augmented Reality*)
  - AR authoring prototyping tool – simple AR viewer to download and place 3d models in runtime in AR scene using an android phone and Google poly API(now deprecated) to search and download 3d models
  - https://drive.google.com/file/d/1W_zBqDN4Gx7Z4iDOMD9k7YsZeTpcubyd/view?usp=drive_link

- Draw Together
  (*Unity3D, Multiuser, Networking*)
  - A multiuser collaboration tool prototyped using unity with a shared whiteboard as well as a feature for collaborative mesh painting tool for designers and creators
  - Shared Whiteboard:
    https://drive.google.com/file/d/1IrNWncilm7qRTrPJ404FKYZOEOFnVMYF/view?usp=drive_link
  - Shared Mesh Painting:
    https://drive.google.com/file/d/1algreQwRDNcNN3f3DRehfG1aMzWMSp-_/view?usp=drive_link

- See Like Me: A Colorblindness Simulator
  (*Unity3D, Augmented Reality*)
  - An AR Color Blindness simulator to showcase different color profiles in AR for different cases of color blindness
  - https://drive.google.com/file/d/1O0U5js5ITcljyOSh6oOBnyz0c93D27Xk/view?usp=drive_link

- Covid19 Visualization
  (*Unity3D, WebGL, REST API*)
  - A 3D visualization of Covid19 Cases as obtained from communication with an open source REST API for the Covid Cases data based on geographical location.
  - https://amit-ghimire.github.io/Covid19/